Progress Check Fill the details in respective boxes

Form	When to use?	Name
$y = mx + c$	Slope=m, Intercept=c are given	
$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$		
	The intercepts are given	Intercept form

Example 5.25 Find the equation of a line which passes through (5,7) and makes intercepts on the axes equal in magnitude but opposite in sign.

Solution Let the x intercept be 'a' and y intercept be '$-a$'.

The equation of the line in intercept form is $\frac{x}{a} + \frac{y}{b} = 1$

$$\Rightarrow \frac{x}{a} + \frac{y}{-a} = 1 \text{ (Here } b = -a)$$

$$\therefore x - y = a \,. \qquad ..(1)$$

Since (1) passes through (5,7)

Therefore, $5 - 7 = a \Rightarrow a = -2$

Thus the required equation of the straight line is $x - y = -2$; or $x - y + 2 = 0$

Example 5.26 Find the intercepts made by the line $4x - 9y + 36 = 0$ on the coordinate axes.

Solution Equation of the given line is $4x - 9y + 36 = 0$

we write it as $4x - 9y = -36$ (bringing it to the normal form)

Dividing by -36 we get, $\frac{x}{-9} + \frac{y}{4} = 1 \qquad ...(1)$

Comparing (1) with intercept form, we get x intercept $a = -9$; y intercept $b = 4$

Example 5.27 A mobile phone is put to use when the battery power is 100%. The percent of battery power 'y' (in decimal) remaining after using the mobile phone for x hours is assumed as $y = -0.25x + 1$

(i) Find the number of hours elapsed if the battery power is 40%.

(ii) How much time does it take so that the battery has no power?

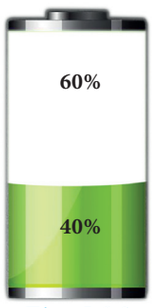


Fig. 5.36

Solution

(i) To find the time when the battery power is 40%, we have to take $y = 0.40$

$$0.40 = -0.25x + 1 \Rightarrow 0.25x = 0.60$$

we get, $x = \frac{0.60}{0.25} = 2.4$ hours.

(ii) If the battery power is 0 then $y = 0$

Therefore, $0 = -0.25x + 1$ gives $0.25x = 1$ hence $x = 4$ hours.

Thus, after 4 hours, the battery of the mobile phone will have no power.

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